

Question	Marking Guidance	Mark	Comments
01.1	A = mitochondrion B = presynaptic membrane/neurone C = (synaptic) vesicle(s) D = synaptic gap/cleft;;	2 (2 x AO1)	4 correct = 2 marks 2–3 correct = 1 mark 0–1 correct = 0 marks A. Accept mitochondria/crista(e). B. Accept synaptic knob/bulb or motor neurone. B. Accept nerve cell for neurone.

Question	Marking Guidance	Mark	Comments
01.2	1. Acetylcholine/neurotransmitter <u>diffuses</u> (across synaptic cleft); 2. (Acetylcholine/neurotransmitter) attaches to <u>receptors</u> on the sarcolemma; 3. Sodium <u>ions</u> enter leading to depolarisation/action potential; 4. Calcium (ions) released by endoplasmic/sarcoplasmic reticulum;	4 (4 x AO1)	Incorrect sequence, penalise one mark. 1 and 2. Accept abbreviations for acetylcholine as term is in the question. 1, 2 and 3. Accept mark points in 'context of a postsynaptic neurone'. 2. Accept postsynaptic <u>membrane</u> for sarcolemma. 3. Accept Na ⁺ for sodium ions. 3. 'Sodium ion channels opening' on its own is not enough. 4. Accept Ca ²⁺ / Ca (ions not required as given in question).

Question	Marking Guidance	Mark	Comments
01.3	1. (Inside postsynaptic neurone/membrane/axon) is more negative OR Membrane/neurone/axon) potential is below resting potential	2 (2 x AO1)	1. Accept answers which refer to a numerical decrease in the resting potential.

	OR Potential difference (across membrane) is greater; 2. More sodium ions (required to enter) for depolarisation OR More sodium ions (required to enter) for action potential OR Prevents sodium ions causing depolarisation;		1. Accept answers which refer to a 'decrease in the resting potential'. 1. Accept answers which refer to 'reduces/lowers potential' but not 'reduces potential difference'. 1. Accept p d for potential difference. 2. Accept to 'reach threshold' or 'generator potential' for 'action potential'. 2. Accept Na⁺ for sodium ions.
--	---	--	--